Formalizing Fefferman's Official Periodic Navier–Stokes Problem (B) in Lean 4: Global Linear Strain Exclusion on $\mathbb{T}^3$, Biot–Savart Direction Depletion Identities, and Conditional Reductions via Beale–Kato–Majda

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Abstract

We present a self-contained Lean 4 formalization (`leanprover/lean4:v4.30.0-rc1`, Mathlib 4, 224 theorem/lemma declarations in source across 7 modules with zero `sorry` and zero non-standard axioms outside `[propext, Classical.choice, Quot.sound]`) of Charles L. Fefferman's official periodic 3D incompressible Navier–Stokes problem statement (*Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute, 2006, Statement (B) on $\mathbb{R}^3/\mathbb{Z}^3$) together with unconditional geometric/kinematic identities, spatial-maximum Bochner–polar reduction, and explicit **conditional reductions** to Beale–Kato–Majda continuation. In `FeffermanClayNS/FinalProductTheorem.lean`, all five unconditionally proved components are packaged into the single 5-conjunct master product theorem `fefferman_periodic_ns_exact_reduction_product_theorem` ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$).

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1 Introduction and Single-Sentence Final Product Form ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$)

In his official Clay Mathematics Institute problem description [1], Charles L. Fefferman formulates the periodic existence and smoothness problem (Statement (B)) for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For the unmodified Navier–Stokes PDE ($\kappa = 0$ in the dynamics), Lean's kernel cannot derive $\langle \omega, (\omega\cdot\nabla)u\rangle + \nu\langle \omega, \Delta\omega\rangle \le C_{\mathrm{CZ}} M^3 - \kappa'' M^4$ at a spatial maximum without either taking the high-set direction-curvature bound $|\nabla\hat{\xi}(x_{\max})|^2 \ge$

$c_{\mathrm{dir}}M$ as an explicit hypothesis or proving that open PDE estimate from scratch. Accordingly, this repository states **only the theorems it proves explicitly and unconditionally**, packaged in `FeffermanClayNS/FinalProductTheorem.lean` into a single 5-conjunct product theorem.

1.1 Single-Sentence English Prose Statement

Theorem 1.1 (Master Product Theorem: `fefferman_periodic_ns_exact_reduction_product_theorem`). *For every viscosity $\nu \in \mathbb{R}$ on the 3-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, **($Q_1$)** spatial $\mathbb{Z}^3$-periodicity is equivalent to invariance under the three unit coordinate shifts $f(x + e_j) = f(x)$, and Fefferman's energy-bounded periodic statement implies his exact periodic statement (*$\mathrm{Clay_B_exact_with_energy}(\nu) \implies \mathrm{Clay_B_exact}(\nu)$*); **($Q_2$)** every $\mathbb{Z}^3$-periodic linear map $L : \mathbb{R}^3 \to \mathbb{R}^3$ vanishes identically ($L = 0$), excluding any non-zero global linear Burgers background strain $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ ($\alpha \neq 0$) on $\mathbb{T}^3$; **(Q_3)** the Biot–Savart vortex-stretching determinant obeys $\det(\omega(y), z, \omega(x)) = \det(\omega(y) - \omega(x), z, \omega(x))$ and vanishes under collinear alignment $\omega(y) = c\,\omega(x)$, while under parabolic rescaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$ both cubic stretching $C_{\mathrm{CZ}}|\omega_\lambda|^3$ and viscous direction-curvature dissipation $\nu|\omega_\lambda|^2|\nabla\hat{\xi}_\lambda|^2$ scale homogeneously as λ^6; **(Q_4)** at a spatial maximum $x_{\max}$ where $\langle\omega, (\omega\cdot\nabla)u\rangle \le C_{\mathrm{CZ}}M^3$, $\nu\langle\omega, \Delta\omega\rangle \le -\nu M^2|\nabla\hat{\xi}|^2$, and high-set direction-curvature coercivity $|\nabla\hat{\xi}(x_{\max})|^2 \ge c_{\mathrm{dir}}M$ holds, the maximum vorticity rate satisfies $M'(t) \le C_{\mathrm{CZ}}M(t)^2 - (\nu c_{\mathrm{dir}})M(t)^3$, which forces the uniform Lyapunov ceiling $M(t) \le \max(M(0), C_{\mathrm{CZ}}/(\nu c_{\mathrm{dir}}))$; and **($Q_5$)** whenever a periodic flow class satisfies the high-set direction-curvature depletion hypothesis* $\mathrm{PeriodicDirectionDepletionBound}(\nu)$ *and classical periodic Beale–Kato–Majda continuation* $\mathrm{PeriodicBKMContinuation}(\nu)$, *Fefferman's periodic statement* $\mathrm{Clay_B_exact}(\nu)$ *holds.*

1.2 Exact Mathematical Product Form ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$)

$$
\begin{aligned}
\mathcal{Q}_{\mathrm{ClayB}}(\nu) \;:\Longleftrightarrow\; & \underbrace{\Big((\forall f : \mathbb{R}^3 \to \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(f) \iff \forall x,\, j \in \{1,2,3\},\ f(x+e_j) = f(x)) \ \wedge\ (\mathrm{Clay_B_exa}\Big)}_{Q_1(\nu)\text{: Exact Fefferman Periodic Unit-Shift Equivalence \& Energy I}} \\
& \wedge \underbrace{\Big((\forall L : \mathbb{R}^3 \to_{\mathbb{R}} \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(L) \implies L = 0) \ \wedge\ (\forall \alpha \neq 0,\ \neg\mathrm{IsZ3Periodic}(\mathrm{diag}(-\tfrac{\alpha}{2}, -\tfrac{\alpha}{2}\Big)}_{Q_2\text{: Torus Exclusion of Global Linear Burgers Strain}} \\
& \wedge \underbrace{\Big(\det(\omega_y, z, \omega_x) = \det(\omega_y - \omega_x, z, \omega_x) \ \wedge\ \det(c\omega_x, z, \omega_x) = 0 \ \wedge\ C_{\mathrm{CZ}}(\lambda^2|\omega|)^3 - \nu(\lambda^2|\omega|)^2(\lambda|}_{Q_3\text{: Biot–Savart Alignment Depletion \& } \lambda^6 \text{ Parabolic Scaling Homogene}} \\
& \wedge \underbrace{\Big((\mathrm{Stretch} \le C_{\mathrm{CZ}}M^3 \wedge \mathrm{Diff} \le -\nu M^2|\nabla\hat{\xi}|^2 \wedge |\nabla\hat{\xi}|^2 \ge c_{\mathrm{dir}}M \implies M' \le C_{\mathrm{CZ}}M^2 - \nu c_{\mathrm{dir}}M^3}_{Q_4\text{: Spatial-Maximum Bochner–Polar Reduction \& C}} \\
& \wedge \underbrace{\Big((\mathrm{PeriodicDirectionDepletionBound}(\nu) \ \wedge\ \mathrm{PeriodicBKMContinuation}(\nu)) \implies \mathrm{Clay_B_}}_{Q_5(\nu)\text{: Exact Conditional Implication to Fefferman's Clay_B_exact}(\nu)}
\end{aligned}
\tag{1}
$$

2 Verbatim Lean 4 Kernel Audit and Cryptographic Attestation

A comment-aware count of `theorem`/`lemma` declaration headers in the Lean sources (the `python3` command in the repository README) gives 224 declarations across 7 modules. `AxiomAudit.lean` audits 231 names (those 224 theorem/lemma declarations together with structure fields and definitions). The titan audit reports more because it counts compiled/generated constants. The master product theorem depends only on the standard axioms `[propext, Classical.choice, Quot.sound]`.

Table 1: Summary of Unconditionally Proved Theorems vs. Open Analytic Hypotheses in `FeffermanClayNS` (224 theorem/lemma declarations, 0 `sorry`).

Declaration / Mathematical Item	Status	Mathematical Meaning & Scope
`fefferman_periodic_ns_exact_reduction_product_theorem` (`FinalProductTheorem.lean`)	**PROVED** (Unconditional, 21 theorem/lemma declarations)	Master 5-conjunct product theorem ($Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu)$) and spatial-maximum Bochner–polar reduction.
`Clay_B_exact` ν, `isZ3Periodic_iff_unit_shifts`, `clay_B_exact_of_with_energy`	**PROVED** (Unconditional)	Exact encoding of Fefferman's Statement (B), unit-shift periodicity equivalence, and logical implication from the energy-bounded variant.
`z3_periodic_linear_eq_zero`, `burgersLinearStrain_not_z3_periodic`	**PROVED** (Unconditional)	Any $\mathbb{Z}^3$-periodic linear map on $\mathbb{R}^3$ vanishes identically; the global linear Burgers strain $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ ($\alpha \neq 0$) is not $\mathbb{Z}^3$-periodic.
`biot_savart_det_aligned_eq_zero`, `aligned_vorticity_stretching_integrand_zero`, `biot_savart_det_difference_identity`	**PROVED** (Unconditional)	Exact pointwise vanishing of $\det(\omega(y), z, \omega(x))$ when $\omega(y) = c\,\omega(x)$, reducing the stretching kernel to direction increments $\omega(y) - \omega(x)$.
`first_remaining_gap_bochner_polar_chain_at_rest`, `scale_invariant_cubic_rate_of_direction_coercivity`, `comparison_ode_ceiling`	**PROVED** (Unconditional)	Spatial-maximum Bochner–polar reduction from $\lvert\nabla\hat{\xi}\rvert^2 \geq c_{\mathrm{dir}}\lvert\omega\rvert$ to $M' \leq C_{\mathrm{CZ}}M^2 - \kappa'' M^3$ and ODE comparison ceiling $M(t) \leq \max(M(0), C_{\mathrm{CZ}}/\kappa'')$.
`clay_B_exact_of_periodic_tether_bound`, `clay_B_exact_of_torus_direction_depletion`	**PROVED CONDITIONAL REDUCTIONS**	Conditional theorems proving that *if* the open direction-depletion/tether hypothesis and BKM continuation hold, then `Clay_B_exact` ν follows.
`periodicTstar_eq_top_of_bkm`, `periodicGlobalOfTstarTop`, `clay_B_exact_of_periodic_bkm`	**PROVED CONDITIONAL REDUCTION**	Smooth solutions on $[0,T]$ carry no $L^2(\mathbb{R}^3)$ clause (it fails for every nonzero periodic field; the earlier definitions, kept with suffix `Original`, made `PeriodicLocalExistence` and `PeriodicMaximalSolution` false, as proved by `not_periodicLocalExistenceOriginal`, `not_periodicMaximalSolutionOriginal`), and maximality is required only for $0 < T$. With $T^* = \sup\{T :$ a periodic smooth solution exists on $[0,T]\}$ in $\overline{\mathbb{R}}$: `PeriodicLocalExistence`, `PeriodicMaximalSolution`, `PeriodicBKMCriterion` (classical, not formalized; the global step `periodicGlobalOfTstarTop` is proved) and `PeriodicBKMIntegrability` (OPEN: $\int_0^T \lVert\omega\rVert_\infty < \infty$ for every T along the maximal solution) $\Rightarrow T^* = \infty \Rightarrow$ `Clay_B_exact`.
`periodicFiniteTimeBound_vacuous`, `periodicFiniteTimeVorticityBound_of_finite`, `periodicBKMContinuation_of_finite_scaffold`, `clay_B_exact_of_periodic_finite_time_bound_scaffold`	**PROVED (SCAFFOLDING)**	The vorticity-bound Props quantify only over solutions already existing on $[0,T]$ and hold with $C = 0$ where none exists, so in these reductions the continuation hypothesis carries all of Statement (B).

Listing 1: Source `theorem`/`lemma` declaration counts in `fefferman-clay-navier-stokes` (not a titan-audit or AxiomAudit total).

```
FeffermanClayNS.Defs: theorems=131
FeffermanClayNS.FeffermanStatements: theorems=10
FeffermanClayNS.FinalProductTheorem: theorems=21
FeffermanClayNS.HighSetDepletionAndBKM: theorems=26
FeffermanClayNS.MainTheorem: theorems=1
FeffermanClayNS.Stage6Calculus: theorems=16
FeffermanClayNS.TorusStrainExclusion: theorems=19
TOTAL theorems=224
'FeffermanClayNS.fefferman_periodic_ns_exact_reduction_product_theorem' depends on axioms: [
    propext, Classical.choice, Quot.sound]
```

A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/fefferman-clay-navier-stokes
- Verified commit:
  `1d5e04eb88727ba0400a0faa8cd3f0b4333d773d`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  https://github.com/BenFrohman/fefferman-clay-navier-stokes/actions/runs/41574094256
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  https://github.com/BenFrohman/fefferman-clay-navier-stokes/archive/1d5e04eb88727ba0400a0faa8cd3f0b4333d773d.tar.gz
  `bd4580341fe2206704b2be63c25d8273894346bd44051f60909b41ab8f3615f9`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. `AxiomAudit.lean` audits the 231 names listed below, of which 224 are theorem or lemma declarations in source; the remaining names are structure fields and definitions. Every one of these 231 names depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`NavierStokes3D.KatoLocalWitness.Tpos`	`[propext, Classical.choice, Quot.sound]`
`NavierStokes3D.KatoLocalWitness.init`	`[propext, Classical.choice, Quot.sound]`
`NavierStokes3D.KatoLocalWitness.pde`	`[propext, Classical.choice, Quot.sound]`
`NavierStokes3D.KatoLocalWitness.phys`	`[propext, Classical.choice, Quot.sound]`

Lean declaration	#print axioms
NavierStokes3D.KatoLocalWitness.smooth	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_iff_forall_lt	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_katoRestartTime	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_unbounded_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_eq_top_of_uniform_restart	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_nonneg	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_pos_of_exists_smooth_solution	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_pos_of_kato_witness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.Tstar_zero_eq_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.pressureC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC1	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spaceC3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.VorticityTransportRegularity.spacetimeC2	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda_scaffold	[propext, Classical.choice, Quot.sound]
NavierStokes3D.beale_kato_majda_scaffold_integrability_unused	[propext, Classical.choice, Quot.sound]
NavierStokes3D.bkm_criterion_without_integrability_not_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.contDiffAt_directionalCoord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_difference	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_eq_sum_directional	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_norm_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_norm_le_of_CZ	[propext, Classical.choice, Quot.sound]
NavierStokes3D.convective_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_add	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_add3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_apply	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_convective	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.curl_convective_div_free	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_gradient	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_laplacian	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_time_deriv	[propext, Classical.choice, Quot.sound]
NavierStokes3D.curl_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.cyclicLiePairing_expand	[propext, Classical.choice, Quot.sound]
NavierStokes3D.deriv_euclidean_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.deriv_prod_fst_slice	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_convective_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_coord_of_differentiableAt_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_curl_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_directionalCoord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_of_contDiffAt_coords	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_pressureGradient_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.differentiableAt_time_deriv_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_convective	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_laplacian	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_pressureGradient	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.directionalCoord_symm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_add_smul	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_of_eq_curl	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_smul	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.div_smul_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.div_zero_field	[propext, Classical.choice, Quot.sound]
NavierStokes3D.energy_zero_of_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.energy_zero_of_gronwall_Icc	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_of_l2_energy_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_zero_of_isPhysicalNSSolution_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.eq_zero_of_isPhysicalNSSolution_zero_all	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ereal_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.exists_mem_existenceTimes_gt_of_lt_Tstar	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_apply_fderiv	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_directionalCoord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_field_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.fderiv_prod_snd_slice	[propext, Classical.choice, Quot.sound]
NavierStokes3D.gradient_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_eq_inner_grad_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_le_strain	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_convective_product_rule	[propext, Classical.choice, Quot.sound]
NavierStokes3D.inner_self_convective_eq_inner_grad_half_norm_sq	[propext, Classical.choice, Quot.sound]
NavierStokes3D.integration_by_parts_of_vanishing_flux	[propext, Classical.choice, Quot.sound]
NavierStokes3D.isPhysicalNSSolution_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.katoRestartTime_pos	[propext, Classical.choice, Quot.sound]
NavierStokes3D.katoSobolevIndex_lt	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_Hs_bound_on_restart_interval	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_ponce_implies_quadratic	[propext, Classical.choice, Quot.sound]
NavierStokes3D.kato_ponce_of_uniform_vorticity	[propext, Classical.choice, Quot.sound]
NavierStokes3D.laplacian_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.laplacian_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.le_Tstar_of_mem_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.le_vorticity_sup_norm	[propext, Classical.choice, Quot.sound]
NavierStokes3D.local_existence	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.mem_existenceTimes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_add_of_kato_restart	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_of_kato_witness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.mem_existenceTimes_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.navier_stokes_eq_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_eq_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_cauchy_on_Icc_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_momentum_difference	[propext, Classical.choice, Quot.sound]
NavierStokes3D.ns_pde_shift	[propext, Classical.choice, Quot.sound]
NavierStokes3D.one_add_log_le_self	[propext, Classical.choice, Quot.sound]
NavierStokes3D.parabolic_regularity_from_vorticity_bound	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressureGradient_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressureGradient_sub	[propext, Classical.choice, Quot.sound]
NavierStokes3D.pressure_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.reciprocal_of_quadratic_growth	[propext, Classical.choice, Quot.sound]
NavierStokes3D.restart_of_kato_quantitative	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_below_Tstar_of_uniqueness	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_of_Tstar_top	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smoothness_on_restart_interval	[propext, Classical.choice, Quot.sound]
NavierStokes3D.smul_coord	[propext, Classical.choice, Quot.sound]
NavierStokes3D.strain_sup_le_of_CZ	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_enstrophy_di_at_spatial_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_inner_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_inner_le_at_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_pairing_le_at_spatial_max	[propext, Classical.choice, Quot.sound]
NavierStokes3D.stretching_quartic_le	[propext, Classical.choice, Quot.sound]
NavierStokes3D.sum_univ_fin3	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_const	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_shift	[propext, Classical.choice, Quot.sound]
NavierStokes3D.time_deriv_sub	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
NavierStokes3D.time_space_mixed_partials	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transport_inner_vanishes_of_grad_zero	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_energy_pairing_vanishes	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_scalar_along_characteristic	[propext, Classical.choice, Quot.sound]
NavierStokes3D.transported_scalar_maximum_principle	[propext, Classical.choice, Quot.sound]
NavierStokes3D.uniqueness_of_existence_times_of_energy_gronwall	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_sup_norm_nonneg	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_at	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_equation	[propext, Classical.choice, Quot.sound]
NavierStokes3D.vorticity_transport_inner	[propext, Classical.choice, Quot.sound]
gradient.congr_simp	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_with_energy	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.isZ3Periodic_iff_unit_shifts	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.library_huniq_unsatisfiable_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_hInt	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_huniq_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.repaired_huniq_shift_at_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.stepField_not_physical	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.vorticity_sup_norm_zero_field	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.AttainedSpatialMaximum.attained	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.AttainedSpatialMaximum.upper_bound	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.AttainedSpatialMaximum.vorticity_sup_norm_eq	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.bddAbove_range_norm_of_attained_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.beltrami_amplitude_pos_and_bounded	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.beltrami_lamb_vector_cross3_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.bochner_laplacian_3d_identity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.concrete_bochner_polar_viscous_damping_at_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_convective_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_isPeriodicSmoothInitialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_isPeriodicSmoothSolution	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.constVelocity_isZ3Periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_laplacian_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constVelocity_ne_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constantin_fefferman_near_far_cutoff_bridge	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.fefferman_periodic_ns_exact_reduction_product_theorem	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.first_remaining_gap_bochner_polar_damping_at_max	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.near_field_stretching_absorbed_by_viscous_direction_damping	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.nonzero_periodic_smooth_solution_exists	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.polar_pythagoras_3d_sum	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.polar_pythagoras_directional_deriv	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unitCube_nonempty	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.vorticity_sup_norm_eq_of_isSpatialMax	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_bkm	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_finite_time_bound_scaffold	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_periodic_tether_bound	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_B_exact_of_torus_direction_depletion	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.comparison_ode_ceiling	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.constE0_initialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cubic_comparison_field_eq	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cubic_comparison_field_neg_of_gt	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.isPeriodicSmoothSolution_of_forall_horizon	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_isPhysicalNSSolution_of_constE0	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_periodicLocalExistenceOriginal	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.not_periodicMaximalSolutionOriginal	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicBKMContinuation_of_finiteTime_scaffold	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicExistenceTimesOriginal_constE0	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicFiniteTimeBound_vacuous	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicFiniteTimeVorticityBound_of_uniform	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicGlobalOfTstarTop	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicTetherClosingBound_of_directionDepletion	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.periodicTstar_eq_top_of_bkm	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.scale_invariant_cubic_rate_of_direction_coercivity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unmodified_max_rate_of_cubic_source	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.viscous_direction_dissipation_bridge	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.zero_isPeriodicSmoothInitialData	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.zero_isPeriodicSmoothSolutionOn	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.clay_reduces_to_direction_curvature_bound	[propext, Classical.choice, Quot.sound]
Stage6.bochner_leibniz_of_hasDerivAt	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_half_norm_sq	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_inner_self	[propext, Classical.choice, Quot.sound]
Stage6.hasDerivAt_polar	[propext, Classical.choice, Quot.sound]
Stage6.inner_deriv_zero_at_max	[propext, Classical.choice, Quot.sound]
Stage6.polar_pythagoras_of_hasDerivAt	[propext, Classical.choice, Quot.sound]
Stage6.real_iSup_eq_zero_of_not_bddAbove	[propext, Classical.choice, Quot.sound]
Stage6.shearU_hasDerivAt_z	[propext, Classical.choice, Quot.sound]
Stage6.shearU_heat	[propext, Classical.choice, Quot.sound]
Stage6.shearV_hasDerivAt_z	[propext, Classical.choice, Quot.sound]
Stage6.shearV_heat	[propext, Classical.choice, Quot.sound]
Stage6.shear_beltrami	[propext, Classical.choice, Quot.sound]
Stage6.shear_convective_and_div_zero	[propext, Classical.choice, Quot.sound]
Stage6.shear_nonconstant	[propext, Classical.choice, Quot.sound]
Stage6.shear_periodic	[propext, Classical.choice, Quot.sound]
Stage6.unit_direction_deriv_orthogonal	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.aligned_vorticity_stretching_integrand_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.biot_savart_det_aligned_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.biot_savart_det_difference_identity	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_coord	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_div_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgersLinearStrain_not_z3_periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.burgers_background_strain_not_z3_periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.cross3_self	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.det3_self_right	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
FeffermanClayNS.linear_strain_not_z3_periodic	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.navier_stokes_scaling_homogeneity_deg6	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.navier_stokes_vorticity_dt_scaling_exponent	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.t3_eq_sum_stdBasisVec	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.unnormalized_quartic_scaling_mismatch	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_continuousLinear_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_coord_diff_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_deriv_integral_unit_interval_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_linear_eq_zero	[propext, Classical.choice, Quot.sound]
FeffermanClayNS.z3_periodic_unit_shift	[propext, Classical.choice, Quot.sound]

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